

Henry Harvin
YUVA INTERN PROGRAM

SQL DATA ANALYST - AGRIBUSINESS

Week 1 Project Report
Data Acquisition, Cleaning & Preliminary Transformation Using SQL

Dataset:

India Crop Production - State Wise

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Introduction

Agriculture is an important part of India's economy. It involves the production of different crops across many states and districts. Since crop production can change depending on the crop, season, location, and year, agricultural data can provide useful information for analysis and decision-making.

For this project, the India Crop Production - State wise dataset has been selected. The dataset contains information about states, districts, crop years, seasons, crops, area of cultivation, and production. It contains 246,091 records and 7 columns, making it suitable for practising data analysis and data cleaning using SQL.

The main focus of this project is to understand and prepare raw agricultural data before it is used for further analysis. In a real data analytics project, raw data may contain problems such as missing values, duplicate records, extra spaces, or data stored in the wrong format. These problems need to be identified and handled before analysing the data.

In this project, the dataset is imported into MySQL and first checked to understand its structure and quality. SQL queries are then used to identify missing values, duplicate records, extra spaces, and formatting issues in the data.

After checking the raw data, a separate cleaned table is created. The cleaning process includes removing unnecessary spaces, converting blank values into NULL where required, and converting area and production values into numeric data types. The original raw table is kept unchanged so that the cleaning process can be compared with the original data.

The project therefore covers the basic steps of a data analytics workflow: data acquisition, data quality checking, data cleaning, preliminary transformation, and validation. The cleaned data can later be used for further analysis of crop production across different states, districts, crops, seasons, and years.

Project Objectives

The main objective of this project is to understand and perform the initial steps of a data analytics project using an agricultural dataset and SQL.

The specific objectives are:

1. To select a suitable public agricultural dataset that contains useful information for data analysis.
2. To understand the structure and contents of the dataset, including its columns, data types, and the type of information stored in each field.
3. To import the dataset into MySQL and store the original data in a raw table.
4. To check the quality of the raw data and identify problems such as missing values, duplicate records, unnecessary spaces, and formatting issues.
5. To analyse the identified data quality issues and understand how they may affect further analysis.
6. To clean the dataset using SQL by removing unnecessary spaces, handling blank values, and converting data into suitable formats.
7. To keep the original raw data unchanged and create a separate cleaned table for the processed data.
8. To validate the cleaned data by comparing it with the original data and checking whether the cleaning process was completed correctly.
9. To document the complete process using SQL queries, explanations, results, and screenshots so that the work can be easily understood and followed by another person.
10. To prepare the dataset for further analysis, such as studying crop production by state, district, crop, season, and year.

Dataset Selection

For this project, the India Crop Production - State wise dataset was selected from Kaggle. The dataset was created by AravindPCoder and is available publicly in CSV format.

The dataset contains 246,091 records and 7 columns. It provides information about crop production in different states and districts of India. The main fields include state, district, crop year, season, crop, area, and production.

This dataset was selected because it is directly related to agriculture, which is the main area of this project. It also contains different types of data, including text, numbers, and dates or years. This makes it useful for practising different SQL operations.

The size of the dataset was another reason for selecting it. With more than 246,000 records, it gives a good opportunity to work with a large dataset instead of a small sample.

During the initial checking of the dataset, some data quality issues were also found. These included missing production values and extra spaces in some text fields. Therefore, the dataset was suitable for demonstrating the process of checking and cleaning raw data using SQL.

The dataset covers crop production records from 1997 to 2015 and includes multiple crops, states, districts, and agricultural seasons. After cleaning, the data can also be used for further analysis based on crop, location, season, and year.

Source

Dataset: India Crop Production - State wise

Source: Kaggle

Creator: AravindPCoder

Format: CSV

Records: 246,091

Columns: 7

Dataset Link - <https://www.kaggle.com/datasets/aravindpcoder/india-crop-production-state-wise>

Dataset Description

The dataset contains information about crop production in different states and districts of India. It has 246,091 rows and 7 columns.

The columns in the dataset are:

Column	Description
State_Name	Name of the state
District_Name	Name of the district
Crop_Year	Year of the record
Season	Season in which the crop was grown
Crop	Name of the crop
Area	Area used for growing the crop
Production	Amount of crop produced

The dataset contains records from 1997 to 2015. It includes different states, districts, crops, and agricultural seasons.

The data contains both text and numerical values. State, district, season, and crop are text fields, while crop year, area, and production are numerical fields.

During the first check, some missing values and extra spaces were found in the data. These issues were later checked and handled during the cleaning process.

Data Acquisition and Import

The dataset was downloaded from Kaggle in CSV format. The file was then imported into MySQL for cleaning and analysis.

First, a database named `agribusiness_week1` was created. A table named `crop_production_raw` was created to store the original data.

The dataset was imported using the load data infile SQL command. During the import, some blank values were found in the Production column. Because of this, the Area and Production columns were temporarily stored as text in the raw table. This helped us import the data without losing the blank values.

The final import was done using the following query:

```
load data infile 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/crop_production.csv'
```

```
into table crop_production_raw
```

```
fields terminated by ','
```

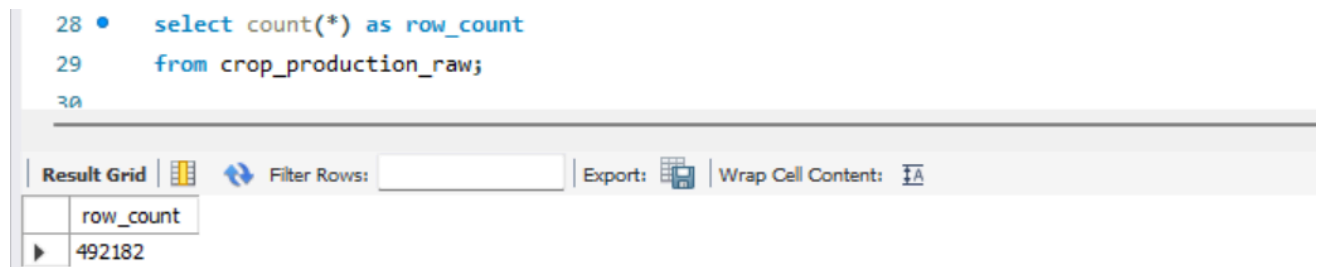
```
enclosed by ''
```

```
lines terminated by '\r\n'
```

```
ignore 1 rows;
```

After importing, the number of rows was checked using SQL. The table contained 246,091 records.

The raw table was kept unchanged so that it could be used for comparison with the cleaned data later.



```
28 • select count(*) as row_count
29     from crop_production_raw;
30
```

row_count
492182

Initial Data Quality Assessment

After importing the dataset, the raw data was checked to find any problems before cleaning it.

The following checks were performed:

- Missing values were checked in all columns.
- Duplicate records were checked.
- Extra spaces in text fields were checked.
- The format of area and production values was checked.
- The crop year range was checked.

The main issues found during the initial check were:

- 3,730 records had missing values in the Production column.
- No missing values were found in the other columns.
- No exact duplicate records were found.
- Extra spaces were found in some state and crop names.
- The Season values had some extra spaces at the end.
- Some area and production values had a different decimal format, such as .20 instead of 0.20.
- The crop years ranged from 1997 to 2015.

These issues were further checked before deciding how to clean the data.

Missing Value Analysis

Missing values were checked in all columns of the dataset.

A total of 3,730 missing values were found in the Production column. No missing values were found in the other columns.

```
36 • select
37     count(*) as total_rows,
38     sum(state_name is null or state_name = '') as missing_state,
39     sum(district_name is null or district_name = '') as missing_district,
40     sum(crop_year is null or crop_year = '') as missing_crop_year,
41     sum(season is null or season = '') as missing_season,
42     sum(crop is null or crop = '') as missing_crop,
43     sum(area is null or area = '') as missing_area,
44     sum(production is null or production = '') as missing_production
45 from crop_production_raw;
```

	total_rows	missing_state	missing_district	missing_crop_year	missing_season	missing_crop	missing_area	missing_production
▶	492182	0	0	0	0	0	0	7460

The missing production values were checked by season and year to understand whether they were limited to a particular group. They were found across different seasons and years.

It was also checked whether the Area value was available when Production was missing. All 3,730 records had an area value available.

```
47 • select
48     count(*) as missing_production,
49     sum(area = '') as missing_area_too,
50     sum(area <> '') as area_available
51 from crop_production_raw
52 where production = '';
```

	missing_production	missing_area_too	area_available
▶	7460	0	7460

Since a missing production value does not mean that the production was zero, these values were not changed to 0. They were kept as missing values and later converted to NULL in the cleaned table.

This allowed the original information to be preserved without making an incorrect assumption about production.

Duplicate Record Analysis

The dataset was checked for duplicate records using SQL.

The total number of records was 246,091. The number of unique records was also 246,091.

This means that no exact duplicate records were found in the dataset.

Since there were no duplicate records, no rows were removed during the cleaning process.

```
54 • select
55     count(*) as total_rows,
56     count(distinct state_name, district_name, crop_year, season, crop, area, production) as unique_rows
57 from crop_production_raw;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	total_rows	unique_rows
▶	492182	246091

Text Formatting Analysis

The text columns were checked for extra spaces because they can cause problems when searching, grouping, or comparing values.

Extra spaces were found in:

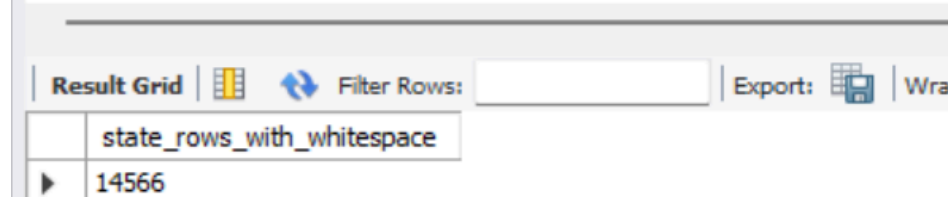
- 14,566 state records
- 1,985 crop records

No extra spaces were found in the district names.

The season values also had extra spaces at the end. The values were checked and found to have the same length, so this was treated as a formatting issue.

The trim() function was used during the cleaning process to remove these extra spaces and make the text values consistent.

```
90 • select
91     count(*) as state_rows_with_whitespace
92   from crop_production_raw
93   where state_name <> trim(state_name);
```



	state_rows_with_whitespace
▶	14566

Numeric Data and Year Validation

The Area, Production, and Crop_Year columns were checked before cleaning.

No missing values were found in the Area column. Some area and production values were written in formats such as .20 and .30 instead of 0.20 and 0.30. These values are still valid numbers, so they were not removed.

The Crop_Year column was also checked to make sure the years were within a reasonable range. The data contains records from 1997 to 2015, with 19 different years.

During cleaning, the Area and Production columns were converted into numeric fields so they could be used correctly for further analysis.

```
160 • select
161     min(crop_year) as earliest_year,
162     max(crop_year) as latest_year,
163     count(distinct crop_year) as unique_years
164 from crop_production_raw;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Co

	earliest_year	latest_year	unique_years
▶	1997	2015	19

Data Cleaning Methodology

After checking the raw data, the identified issues were handled using SQL.

A separate table named `crop_production_cleaned` was created. The original `crop_production_raw` table was kept unchanged.

The following cleaning steps were performed:

- Extra spaces were removed from the state, season, and crop names using `trim()`.
- Blank Area values were converted to NULL.
- Blank Production values were converted to NULL.
- Area and Production were changed from text to numeric values.
- No rows were deleted because there were no duplicate records.
- Missing production values were not changed to zero because missing data does not mean zero production.

```
181 • ○ insert into crop_production_cleaned (  
182     state_name,  
183     district_name,  
184     crop_year,  
185     season,  
186     crop,  
187     area,  
188     production  
189 )  
190 select  
191     trim(state_name),  
192     trim(district_name),  
193     crop_year,  
194     trim(season),  
195     trim(crop),  
196     nullif(trim(area), ''),  
197     nullif(trim(production), '')  
198 from crop_production_raw;  
199
```

The cleaned data was then checked to make sure that the records were not lost and that the cleaning was completed correctly.

Preliminary Transformations

After cleaning the data, some basic transformations were performed to make it easier to use for further analysis.

The following transformations were done:

- Extra spaces were removed from text values.
- Blank values in Area and Production were converted to NULL.
- Area was converted into a numeric value.
- Production was converted into a numeric value.
- The original raw table was kept separately from the cleaned table.

These transformations made the data more consistent and ready for further analysis using SQL or other data analysis tools.

Post-Cleaning Validation

After cleaning the data, the cleaned table was checked to make sure that the process was completed correctly.

The following checks were performed:

- The raw and cleaned tables both contained 246,091 records, so no records were lost.

```
202 • select
203     (select count(*) from crop_production_raw) as raw_rows,
204     (select count(*) from crop_production_cleaned) as cleaned_rows;
```

	raw_rows	cleaned_rows
▶	246091	246091

- The cleaned data still had 3,730 missing production values, which were stored as NULL.
- No extra spaces remained in the state, season, and crop fields.
- All area values were available as numeric values.
- There were 242,361 non-missing production values.

```
228 • select
229     count(*) as total_rows,
230     count(area) as area_values,
231     count(production) as production_values
232 from crop_production_cleaned;
```

	total_rows	area_values	production_values
▶	246091	246091	242361

- The cleaned data contained 33 states, 646 districts, 19 years, 6 seasons, and 124 crops.

```
235 • select
236     count(distinct state_name) as states,
237     count(distinct district_name) as districts,
238     count(distinct crop_year) as years,
239     count(distinct season) as seasons,
240     count(distinct crop) as crops
241 from crop_production_cleaned;
```

	states	districts	years	seasons	crops
▶	33	646	19	6	124

These checks confirmed that the data was cleaned successfully while keeping the original number of records.

Challenges Faced During Data Import and Cleaning

A few problems were faced while importing and cleaning the dataset.

- The MySQL Workbench import option did not complete the full import properly.
- The CSV file used Windows-style line endings, so the correct line ending had to be used in the SQL import command.
- Some Production values were blank. This caused an error when the column was set as a numeric field.
- To solve this, Area and Production were temporarily stored as text in the raw table.
- Some values such as .20 and .30 had a different number format. These were valid numbers, so they were not removed.
- Extra spaces were found in some state, season, and crop values. These were removed using trim().

These issues were handled during the import and cleaning process, and the final cleaned table was successfully created without losing any records.

Final Outcome

After completing the data cleaning process, a clean and structured table was created from the original raw data.

The final outcome was:

- 246,091 records were retained.
- 3,730 missing production values were kept as NULL.
- No duplicate records were found or removed.
- Extra spaces were removed from the required text fields.
- Area and Production were converted into numeric values.
- The raw data was kept separately for reference.
- The cleaned data contains 33 states, 646 districts, 19 years, 6 seasons, and 124 crops.

The cleaned dataset is now ready for further analysis and can be used for studying crop production based on different states, districts, crops, seasons, and years.

Future Scope

The cleaned dataset can be used for further analysis in the future. Some possible areas of analysis are:

- Compare crop production between different states and districts.
- Study crop production trends over different years.
- Compare production across different seasons.
- Find crops with the highest and lowest production.
- Analyse the relationship between cultivation area and production.
- Create charts and dashboards using tools such as Power BI.
- Use the data for further statistical or machine learning analysis.

This project can therefore be used as a starting point for more detailed analysis of agricultural data.

Conclusion

This project focused on the initial steps of working with an agricultural dataset using SQL. The India Crop Production - State wise dataset was downloaded, imported into MySQL, checked for data quality issues, and cleaned.

During the process, missing values, duplicate records, extra spaces, and number formatting were checked. The identified issues were handled using SQL without removing any records unnecessarily.

A separate cleaned table was created while keeping the original raw data unchanged. The final checks showed that all 246,091 records were retained and the cleaned data was ready for further analysis.

Overall, this project helped demonstrate the basic process of data acquisition, data quality checking, data cleaning, transformation, and validation using SQL.

Appendix: Important SQL Queries

The following SQL queries were used during the data import, checking, cleaning, and validation process.

Checking the Number of Records

```
select count(*) as row_count
from crop_production_raw;
```

This was used to check the total number of records after importing the dataset.

Checking Missing Values

```
select
  count(*) as total_rows,
  sum(state_name is null or state_name = "") as missing_state,
  sum(district_name is null or district_name = "") as missing_district,
  sum(crop_year is null or crop_year = "") as missing_crop_year,
  sum(season is null or season = "") as missing_season,
  sum(crop is null or crop = "") as missing_crop,
  sum(area is null or area = "") as missing_area,
  sum(production is null or production = "") as missing_production
from crop_production_raw;
```

This query was used to find missing values in each column.

Checking Duplicate Records

```
select
  count(*) as total_rows,
  count(distinct state_name, district_name, crop_year,
    season, crop, area, production) as unique_rows
from crop_production_raw;
```

This was used to check whether any exact duplicate records were present.

Checking Extra Spaces

```
select
  count(*) as rows_with_trailing_spaces
from crop_production_raw
where crop <> trim(crop);
```

Similar checks were performed for state and district names. This helped identify unnecessary spaces in text fields.

Checking Crop Year

```
select
  min(crop_year) as earliest_year,
  max(crop_year) as latest_year,
  count(distinct crop_year) as unique_years
from crop_production_raw;
```

This was used to check the range of years present in the dataset.

Creating the Cleaned Table

```
create table crop_production_cleaned (
  state_name varchar(100),
  district_name varchar(100),
  crop_year int,
  season varchar(50),
  crop varchar(150),
  area decimal(15,2),
  production decimal(18,2)
);
```

A separate table was created so that the original raw data could be kept unchanged.

Cleaning and Transforming the Data

```
insert into crop_production_cleaned (
  state_name,
  district_name,
  crop_year,
  season,
  crop,
  area,
  production
)
select
  trim(state_name),
  trim(district_name),
  crop_year,
  trim(season),
  trim(crop),
  nullif(trim(area), ""),
  nullif(trim(production), "")
from crop_production_raw;
```

This query removed extra spaces, converted blank values to NULL, and inserted the cleaned data into the new table.

Comparing Raw and Cleaned Records

```
select
  (select count(*) from crop_production_raw) as raw_rows,
  (select count(*) from crop_production_cleaned) as cleaned_rows;
```

This was used to make sure that no records were lost during cleaning.

Final Validation

```
select
  count(*) as total_rows,
  count(area) as area_values,
  count(production) as production_values
from crop_production_cleaned;
```

This was used to check the number of available numeric values after cleaning.

Checking Distinct Values

```
select
  count(distinct state_name) as states,
  count(distinct district_name) as districts,
  count(distinct crop_year) as years,
  count(distinct season) as seasons,
  count(distinct crop) as crops
from crop_production_cleaned;
```

This was used as a final check of the cleaned dataset.